f(x1)=f(x2) uses to proof function is 1-1 then how we say that one to one don't have same output on different inputs

Here is the idea in **very easy words**:

✅ 1. Why do we use $f(x_1) = f(x_2)$ to prove a function is one-to-one?

Because **one-to-one means**:

> If two inputs are different → their outputs must also be different.

This is written as:

> If $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$.

But in mathematics, we usually prove it using the **contrapositive method** (reverse logic), which is easier:

> If $f(x_1) = f(x_2)$, then it must force $x_1 = x_2$.

If this is true, then the function is one-to-one.

✅ 2. Why do we say "one-to-one means different inputs cannot give same output"?

Because the definition of 1-1 function is:

> No two different inputs should have the same output
> (each input has its own unique output).

Example:

f(x)=2x

If 2 inputs give same output:

$2x_1 = 2x_2$ → divide by 2

$x_1 = x_2$

So **no two different x can produce same y**, so it's 1-1.

⭐ VERY SIMPLE SUMMARY

We check one-to-one like this:

1. Assume $f(x_1) = f(x_2)$.
2. Solve the equation.
3. If you always get $x_1 = x_2$ → function is **1-1**.
4. This means: **different inputs can NEVER give the same output**.

Example to make it super clear

$f(x) = x^2$

Check 1-1:

Assume $f(x_1) = f(x_2)$:

$x_1^2 = x_2^2$

→ $x_1 = x_2$ OR $x_1 = -x_2$

Not always $x_1 = x_2$ → so NOT one-to-one.

Because:

$f(2) = 4$

$f(-2) = 4$

Two different inputs → same output → NOT 1-1.

If you want, I can also explain **onto** using the same simple style.

Means actually we want to proof that we input 2 different inputs but output shouldn't be same

Exactly **YES** — you understood it perfectly.
Let me say it in the clearest possible way:

✅ What we *actually* want to prove for 1-1 (injective) function:

"If we put two different inputs, the outputs MUST be different."

Mathematically:

If $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

✅ Why do we use the method $f(x_1) = f(x_2)$?

Because proving the above directly is sometimes difficult.

So instead, we prove the same idea using **reverse logic (contrapositive)**:

> If the outputs are the same → the inputs must also be the same.

This is easier to check.

So we assume:

$f(x_1) = f(x_2)$
And then we show:

→ $x_1 = x_2$

If this is true, then the function is one-to-one.

⭐ SIMPLE EXPLANATION

Saying **f is one-to-one** means:

❌ Two different inputs can *never* give the same output.

✔ Each input has its own unique output.

Example

$f(x) = 3x + 1$

Let $f(x_1) = f(x_2)$:

$3x_1 + 1 = 3x_2 + 1$

$3x_1 = 3x_2$

$x_1 = x_2$

This proves **no two different x can give the same y**, so it's 1-1.

Example of NOT 1-1

$f(x) = x^2$

Let f(2) and f(−2):

Both = 4
Different inputs → same output → NOT one-to-one.

If you want, I can also explain **onto** again in the same easy style.